

Supplementary Materials

Supplementary Materials I: RM-ANOVA results from Foody et al. (2013, 2015)

The paragraphs of results for the primary outcome measures are presented below (emphasis added).

Foody et al. (2013)

Discomfort

“Both conditions recorded similarly low levels of discomfort ($<11/100$) at baseline and increased to the same point after the distress induction (distinction self as context: $+12.7$; hierarchical self as context: $+16.3$; see Figure 1). Thereafter, **distinction resulted in a very small increase in discomfort ($+0.76$), while hierarchy resulted in a decrease (-7.57)**. A mixed between within 3×2 ANOVA revealed a highly significant main effect for time [Wilkes Lambda = $.53$, $F(2, 33) = 14.58$, $p = .000$, partial eta squared = $.47$], but not for condition [$F(1, 34) = 1$, $p = .32$, partial eta squared = $.03$]. **The interaction effect was also non-significant [Wilkes Lambda = $.95$, $F(2, 33) = .81$, $p = .45$, partial eta squared = $.05$]**. Two dependent t-tests investigated which time point was influencing the significant effect. The results showed a significant increase in discomfort from baseline to post-distress induction ($p = .000$), but not from post-distress induction to post-intervention ($p = .31$).” (p. 381)

Anxiety

“Both conditions recorded similarly low levels of anxiety ($<10/100$) at baseline, and both increased at post-distress induction (distinction: $+8.06$; hierarchy: $+13.57$; see Figure 2). **Anxiety subsequently decreased for both conditions, although the larger change was recorded for the hierarchical intervention (distinction: $-.03$; hierarchy: -3.86)**. A mixed between within 3×2 ANOVA revealed a highly significant main effect for time [Wilkes Lambda = $.54$, $F(2, 33) = 14.12$, $p = .000$, partial eta squared = $.46$], but not for condition [$F(1, 34) = .021$, $p = .89$, partial eta squared = $.00$]. **The interaction effect was non-significant [Wilkes Lambda = $.94$, $F(2, 33) = .114$, $p = .33$, partial eta squared = $.07$]**. Two dependent t-tests showed a significant increase in anxiety from baseline to post-distress induction ($p = .000$), but not from post-distress induction to post-intervention ($p = .46$).” (p. 381)

Stress

“Both conditions recorded low stress ($<11/100$) at baseline, and both increased at post-induction (distinction: $+7.43$; hierarchy: $+9.48$; see Figure 3). Thereafter, however, **distinction resulted in an increase in stress ($+4.71$), while hierarchy reduced stress (-8.82)**. A mixed between within 3×2 ANOVA revealed a highly significant main effect for time [Wilkes Lambda = $.65$, $F(2, 33) = 8.74$, $p = .001$, partial eta squared = $.35$], but not for condition [$F(1, 34) = .1$, $p = .76$, partial eta squared = $.00$]. However, **the interaction was significant [Wilkes Lambda = $.83$, $F(2, 33) = 3.46$, $p = .04$, partial eta squared = $.17$]**. Two dependent t-tests showed a significant increase in stress from baseline to post-distress induction for both conditions (both $ps < .05$). The increase in distinction from post-distress induction to post-intervention was not significant ($p = .23$), but the decrease for hierarchy at the same time point was significant ($p = .02$).” (p. 382)

Foody et al. (2015)

Discomfort

A mixed between-within 6×4 ANOVA revealed a highly significant main effect for time, Wilkes Lambda = $.34$, $F(5, 40) = 15.24$, $p = .000$, partial eta squared = $.66$, that accounted

for a large amount of the variance (66 %), but not for condition, $F(3, 44)=.54$, $p=.66$, partial eta squared=.04. In addition, **the interaction effect was not significant, Wilkes Lambda = .88, $F(5, 42)=.58$, $p=.88$, partial eta squared=.07.**

[collapsing across self and object conditions]

A mixed between–within subjects 6 x 2 analysis of variance was conducted to assess the impact of hierarchical and distinction interventions on discomfort ratings across the six time points. This revealed a highly significant main effect for time, Wilkes Lambda=.35, $F(5, 42) = 15.80$, $p=.000$, partial eta squared=.65, that accounted for a large amount of the variance (65 %), but not for condition, $F(1, 46)=.35$, $p = 1.17$, partial eta squared=.03. In addition, **the interaction effect was not significant, Wilkes Lambda = .97, $F(5, 42)=.25$, $p=.94$, partial eta squared=.03.**

Anxiety

A mixed between–within 6 x 4 ANOVA revealed a highly significant main effect for time, Wilkes Lambda=.43, $F(5, 40) = 10.44$, $p=.000$, partial eta squared=.57, that accounted for a substantial amount of the variance (57 %), but not for condition, $F(3, 44)=.77$, $p=.52$, partial eta squared=.05, **or the interaction effect, Wilkes Lambda=.71, $F(5, 42) = 1$, $p=.47$, partial eta squared=.11.**

[collapsing across self and object conditions]

For anxiety ratings between the hierarchical and distinction conditions, a mixed between–within subjects analysis of variance revealed a highly significant main effect for time, Wilkes Lambda=.44, $F(5, 42) = 10.67$, $p=.000$, partial eta squared=.56, but not for condition, $F(1, 46) = 2.04$, $p=.16$, partial eta squared=.04. In addition, **the interaction effect was not significant, Wilkes Lambda=.93, $F(5, 42)=.65$, $p=.67$, partial eta squared=.07.**

Stress

A mixed between–within 6 x 4 ANOVA revealed a highly significant main effect for time, Wilkes Lambda=.52, $F(5, 40) = 7.5$, $p=.000$, partial eta squared=.48, that accounted for a substantial amount of the variance (48 %), but not for condition, $F(3, 43) = 2.14$, $p=.11$, partial eta squared=.12. **The interaction effect was not significant, Wilkes Lambda=.56, $F(5, 42) = 1.71$, $p=.059$, partial eta squared=.17.**

[collapsing across self and object conditions]

For stress ratings between the hierarchical and distinction conditions, a mixed between–within subjects analysis of variance revealed a highly significant main effect for time, Wilkes Lambda=.55, $F(5, 42) = 6.99$, $p=.000$, partial eta squared=.45. Additionally, the main effect for condition was significant, $F(1, 46) = 4.98$, $p=.03$, partial eta squared=.10. **The interaction effect was not significant, Wilkes Lambda=.90, $F(5, 42)=.93$, $p=.47$, partial eta squared=.10.**

Supplementary Materials II: Examples of how Foody et al. (2013, 2015) are cited

Oxford Handbook of Acceptance and Commitment Therapy

Chapter on Cognitive Defusion (Ruiz et al., 2021)

“Foody, Barnes-Holmes, Barnes-Holmes, and Luciano (2013) found that Defusion II [hierarchy] was more efficacious in reducing experimentally induced emotional distress than Defusion I [distinction].”

- Incorrect based on reanalysis

“Similarly, Foody, Barnes-Holmes, Barnes-Holmes, Rai, and Luciano (2015) found that protocols that included framing ongoing private events through hierarchical relations were more efficacious than those that only introduced deictic relations.” (Ruiz et al., 2021, p. 13)

- Incorrect characterisation of results which were in fact null.

Chapter on Clinical behavior analysis and RFT: Conceptualizing psychopathology and its treatment (Luciano et al., 2022)

“Evidence of hierarchically framing ongoing behavior as a central relational process ... Luciano et al. (2011) analyzed the differential effect of two defusion protocols [distinction vs. hierarchy]. ... The results showed that the second protocol had a greater effect on reducing the frequency of problematic behaviors and psychological inflexibility at the 4-month follow-up. Some experimental analogs have replicated the results of the previous study with different dependent variables: experimentally induced emotional distress (Foody, Barnes-Holmes, Barnes-Holmes, & Luciano, 2013)”

- Cites Foody et al. 2013 as evidence of the relevance of RFT to ACT
- Does not acknowledge the failed replication in Foody et al. 2015

Chapter on A primer on Relational Frame Theory (Harte & Barnes-Holmes, 2022)

“Early research in RFT demonstrated a number of distinct patterns of AARR or relational frames. These patterns included: coordination ..., distinction, ... opposition, ... comparison, ... temporality, ... hierarchy (e.g., Foody et al., 2013)” (p.20)

- Mischaracterises nature of original study. Foody et al 2013 did not demonstrate this patter of AARRing but rather assumed it to construct psychological interventions
- Does not cite the failed replication by Foody et al. 2015.

“Behavior Therapy”’s chapter on The Future of Third Wave Cognitive Behavior Therapies (Zettle & Masuda, 2022)

“Illustrative applications of this facet of the CBS model advocated by Hayes et al. (2012a) included efforts by RFT to account for and enhance the use of metaphors (Foody et al., 2014; Sierra et al., 2016) and exercises promoting a type of perspective taking, known as self-as-context (Foody et al., 2013, 2015) within the practice of ACT.” (Zettle and Masuda, 2022, p. 786)

- Doesn't mention whether the efforts were successful or not

Other publications

Approving of Foody

(Kelly & Kelly, 2021)

“The relevance of RFT to ACT is well established in the literature ..., where RFT is explained as having provided the conceptual foundations for ACT (Zettle, 2005) and has made important contributions to an ACT perspective of psychopathology ... and intervention (Foody et al., 2013)” (Kelly and Kelly, 2021, p. 44).

- Cites Foody et al. 2013 as evidence of the relevance of RFT to ACT
- Does not acknowledge the failed replication in Foody et al. 2015

(Dixon & Hayes, 2022)

“This is a concrete example of the way that middle-level terms based on interpretive and empirical functional analyses available at the time can lead to additional basic research that in turn alters the precise meaning of these middle-level terms. For example, self as context interventions based on frames of distinction such as “I am not that” have been found to be generally less effective than interventions based on hierarchical framing such as “I include that” (Foody et al., 2013; Foody et al., 2015).” (Dixon and Hayes, 2022, p. 4)

- Cites Foody et al. 2013 as evidence of the relevance of RFT to ACT
- Cites the failed replication in Foody et al. 2015 as if it was successful

(Law & Hayes, 2021)

“Understanding how to foster healthy perspective-taking seems central to establishing self-direction, independence, and values-based actions (Foody et al., 2015).” (Law and Hayes, 2021, p. 28)

- Mischaracterises the results of the failed replication as if they're positive, and generally mischaracterises the claim which had nothing to do with self-direction, independence, or values-based actions

(Gomide et al., 2024)

“Such an attempt already successfully allowed increasing precision in defusion exercises, in which hierarchical relations with the deictic self-yield a more powerful transfer of function effect compared to other frames (e.g., Foody et al., 2013).” (Gomide et al., 2024, p. 7)

- Incorrect in light of current evidence
- Does not acknowledge the failed replication in Foody et al. 2015

Critical of Foody

(Godbee & Kangas, 2022)

“Although the differences between hierarchical SAC (“I am more than my experiences”) and distinction SAC (“I am not my experiences”) have been researched, there is limited evidence that one type is more effective than the other (Atkins & Styles, 2016; Foody et al., 2013; Foody et al., 2015).” (Godbee and Kangas, 2022, p. 559)

- Arguably could have been more explicit that the effect failed to replicate, but still good to see.
- Unsurprising that more cautious language comes from non ACT insiders and is published in non ACT journals.

References

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